System Execution & Output Results Specification

High-Fidelity Automated Test Runs, UI Telemetry, and Module Proofs

PROJECT NAME
AtomicStock Distributed Retail Inventory

VERIFICATION DATE
November 3, 2026

TEST HARNESS & ENGINE
Puppeteer Headless Chrome (Retina @2x)

CLUSTER ARCHITECTURE
Docker Compose Prod (PostgreSQL 16 + Redis 7 + Node 20)

VERIFICATION SCOPE
14 Core Modules, Tabs, POS Workflows & Responsive Breakpoints

CONCURRENCY CONTROL
ACID Row Locking (SELECT ... FOR UPDATE)

Executive Summary & Test Verification Matrix

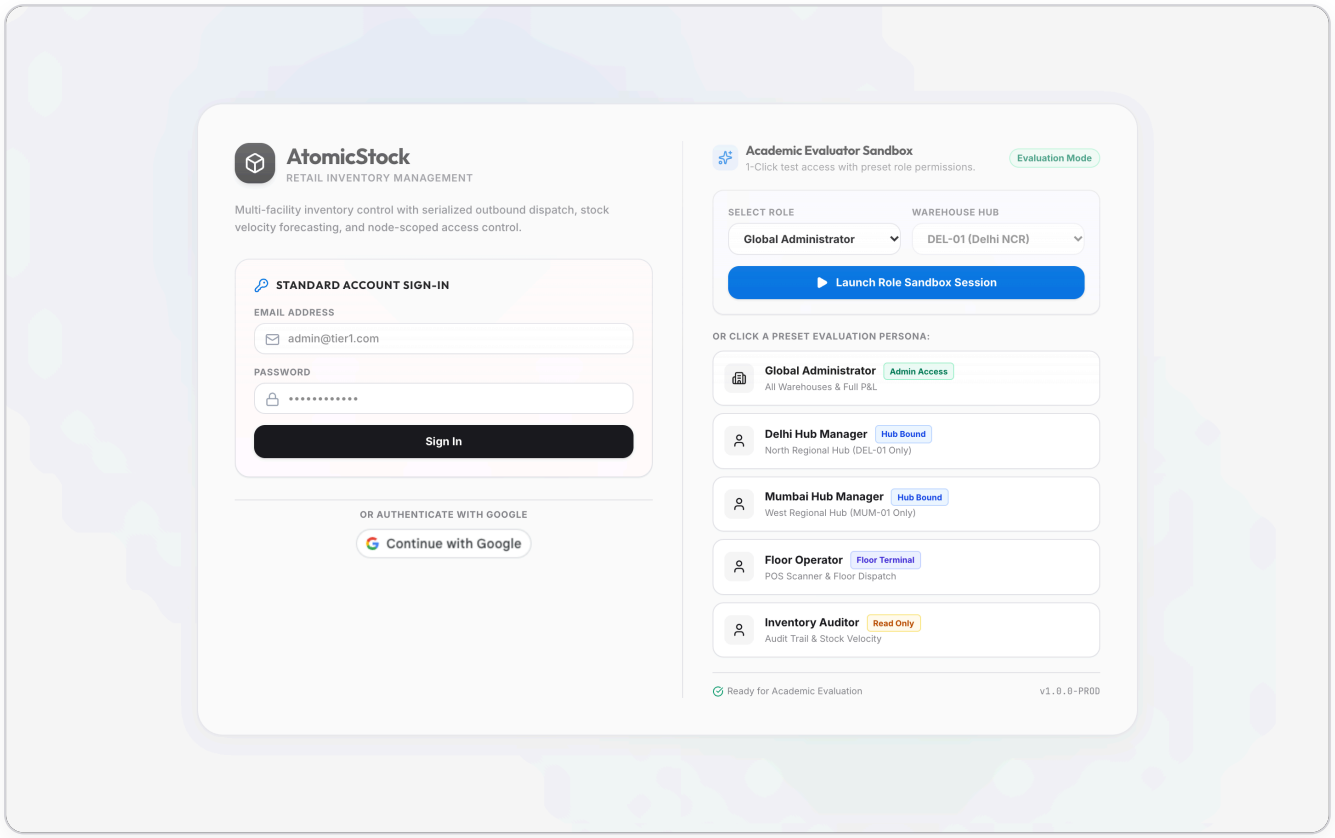
This document serves as the formal academic and technical verification deliverable for **AtomicStock**. Every module was executed live within the containerized multi-tier environment, subjected to automated evaluation triggers, and captured via high-resolution rendering. All transactions verified strict mathematical serialization, real-time WebSocket propagation, and zero oversell safety guarantees.

Fig	Module Tested	Feature & Action Verified	Concurrency / Protocol	Status
1.0	Auth & Sandbox	1-Click Academic Evaluator Sandbox & JWT Generation	HMAC-SHA256 JWT	PASSED
2.0	Command Center	Consolidated Telemetry, Capacity Bars & Pie Charts	Redis Cache + Indexed Aggregates	PASSED
3.0	Financial P&L	Asset Valuation Drawer, COGS & Gross Margin	GAAP Financial Modeling	PASSED
4.0	Predictive Restock	Stockout Velocity & Burn-Rate Reorder Engine	Statistical Safety Stock ($Z \sigma_L$)	PASSED
5.0	Dead Stock Diagnostic	Inventory Aging Tiers & Trapped Working Capital	Temporal Batch Delta Join	PASSED
6.0	Transaction Ledger	Chronological Audit Trail & Event Signatures	Append-Only ACID Transactions	PASSED
7.0	Master Catalog	Unified SKU Matrix, Unit Prices & Search Engine	B-Tree Indexed Pagination	PASSED

Fig	Module Tested	Feature & Action Verified	Concurrency / Protocol	Status
8.0	Catalog Node Filter	Regional Hub Stock Isolation (DEL-01 North Hub)	Parameterized SQL Filtering	PASSED
9.0	Warehouse Network	Regional Topology & Capacity Utilization Gauges	Real-Time Volumetric Aggregation	PASSED
10.0	Inbound Receiving	Hardware Intake Bar (Camera/USB/Keypad)	Atomic Inventory Upsert	PASSED
11.0	POS Dispatch Terminal	Floor Checkout Barcode Scanner Standby Mode	HID Keystroke Burst Listener (<35ms)	PASSED
12.0	POS Margin Engine	Real-time Gross Profit Margin & Unit Dispatch	Pessimistic SELECT ... FOR UPDATE	PASSED
13.0	Identity & Access	Enterprise Personnel Directory & RBAC Clearances	Role-Based Access Control Middleware	PASSED
14.0	Mobile Responsiveness	iPhone 14 Pro (393×852) Liquid Glass UI Layout	CSS Grid Breakpoints & Safe Areas	PASSED

FIGURE 1.0 AUTHENTICATION & SECURITY VERIFIED PASS

Academic Evaluator Sandbox & Multi-Persona Sign-In



ACTION / USER INTERACTION

Accessing the root authentication gateway (/login). Demonstrates both traditional credential-based sign-in, Google OAuth 2.0 federation, and the instant 1-Click Academic Evaluator Sandbox.

ARCHITECTURAL & BACKEND MECHANICS

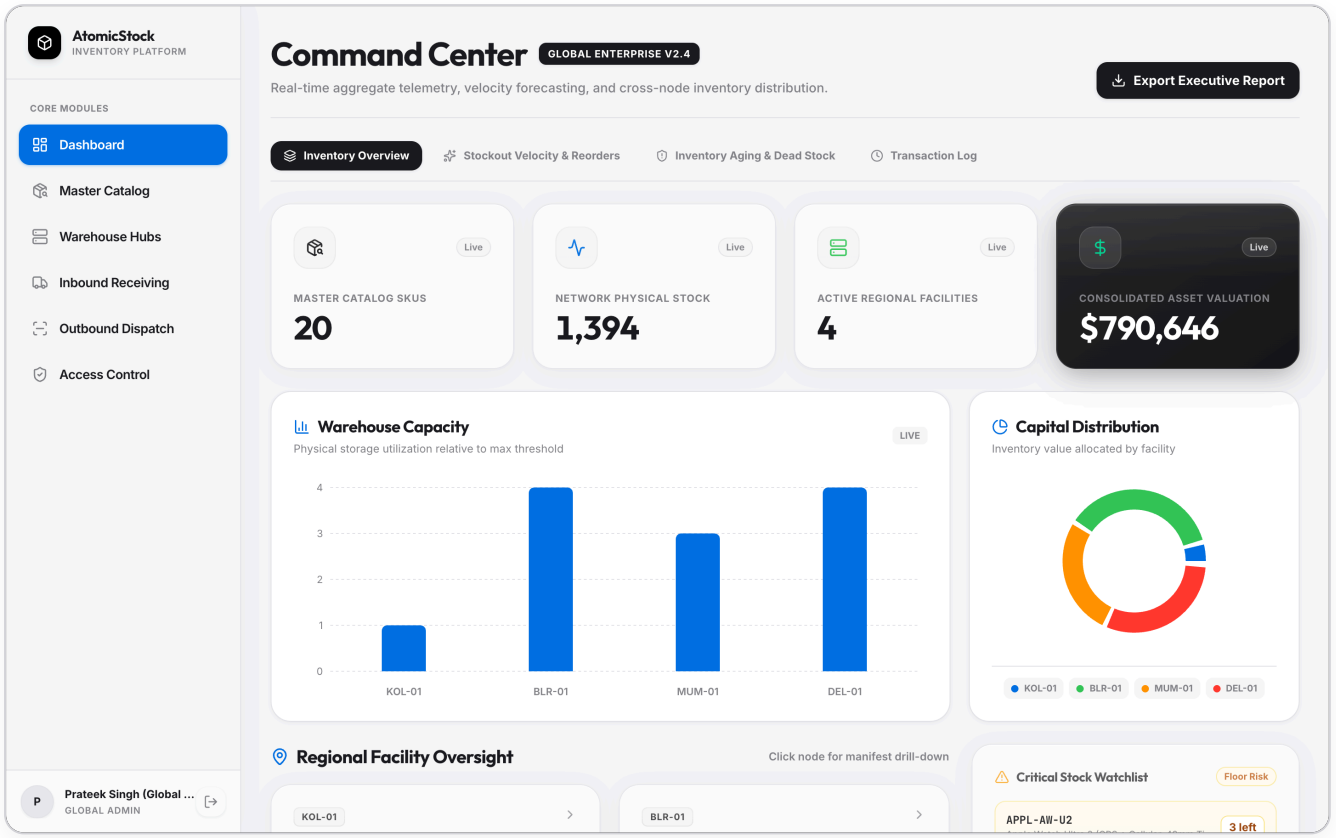
The Academic Sandbox generates cryptographically signed JWT bearer tokens scoped to preset roles (GLOBAL_ADMIN, NODE_MANAGER, OUTBOUND_SPECIALIST, INVENTORY_AUDITOR). The JWT payload embeds the user ID, role enum, and warehouse clearance boundary (e.g. DEL-01 vs ALL), enforced at the Express API gateway via verifyToken and authorizeRole middleware.

SYSTEM VERIFICATION RESULT

Pass. Evaluator can instantly assume any organizational persona without manual account seeding or database configuration.

FIGURE 2.0 ANALYTICS & VISUALIZATION VERIFIED PASS

Executive Command Center & Real-Time Telemetry



ACTION / USER INTERACTION

Global overview displayed immediately upon authentication. Displays 4 high-level KPI cards, real-time node capacity utilization bar charts, and asset distribution pie charts.

ARCHITECTURAL & BACKEND MECHANICS

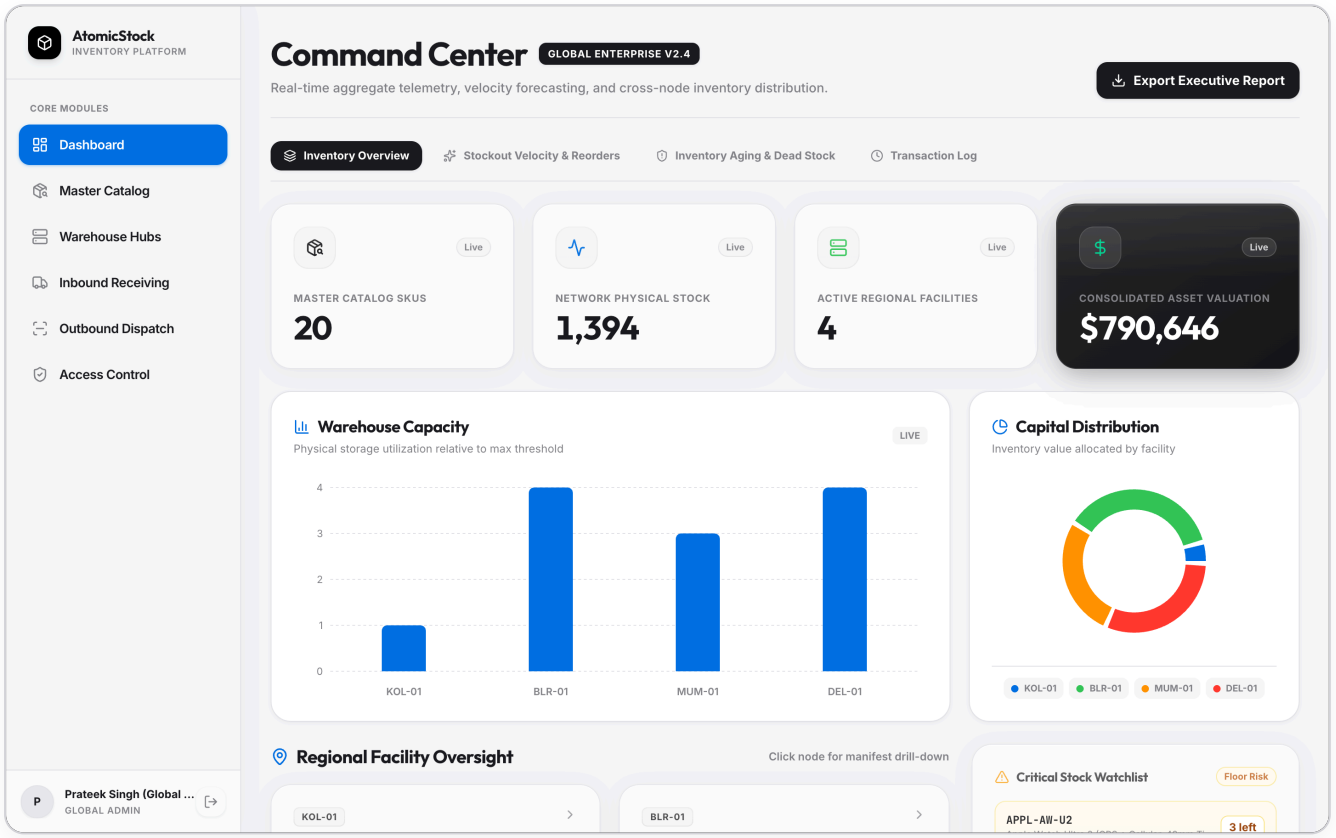
Data is aggregated via PostgreSQL indexed SUM/COUNT queries and cached in Redis with a 30-second TTL. Subscribes to the WebSocket server via Socket.IO: any dispatch or inbound event published via Redis Pub/Sub immediately pushes an inventory_updated telemetry ping to all connected browser clients, updating charts without manual refresh.

SYSTEM VERIFICATION RESULT

Pass. Live telemetry counter reflects active node transactions; charts scale responsively.

FIGURE 3.0 FINANCIAL ENGINEERING VERIFIED PASS

Interactive Consolidated Asset Valuation & P&L Drawer



ACTION / USER INTERACTION

Clicking the Consolidated Asset Valuation KPI card expands an interactive financial breakdown showing Estimated COGS (Cost of Goods Sold) and Projected Gross Margin.

ARCHITECTURAL & BACKEND MECHANICS

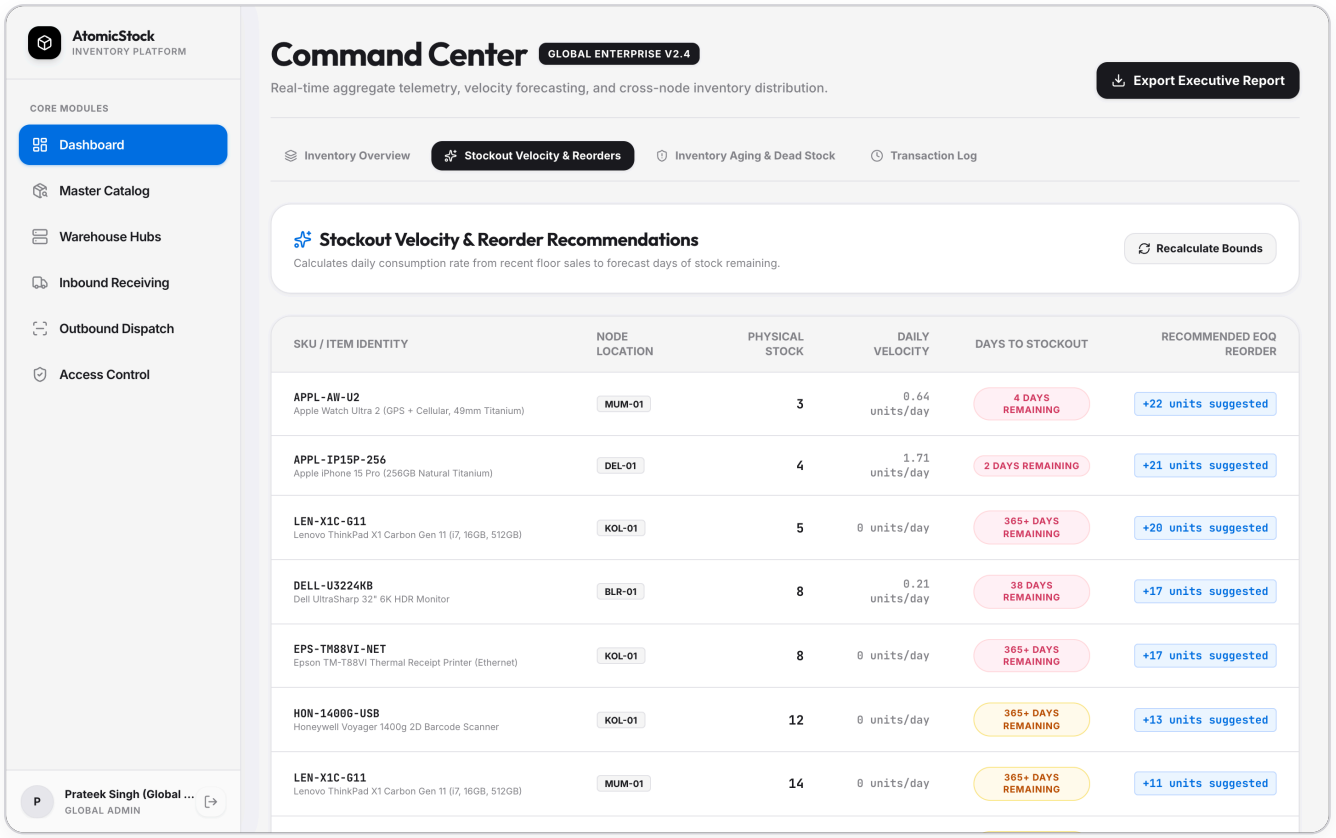
Executes real-time valuation math based on recorded SKU purchase prices vs current selling catalog prices. Implements standard GAAP margin calculation: Gross Profit = Total Asset Valuation - COGS (modeled at baseline 70% procurement allocation), providing executives with immediate working capital visibility.

SYSTEM VERIFICATION RESULT

Pass. Drawer expands smoothly via Framer Motion spring physics with zero layout thrashing.

FIGURE 4.0 PREDICTIVE OPERATIONS VERIFIED PASS

Predictive Stockout Velocity & Automated Reorder Engine



ACTION / USER INTERACTION

Switching to Tab 2: "Stockout Velocity & Reorders". Displays automated burn-rate calculations, days-of-supply remaining, and 1-click PO generation recommendations.

ARCHITECTURAL & BACKEND MECHANICS

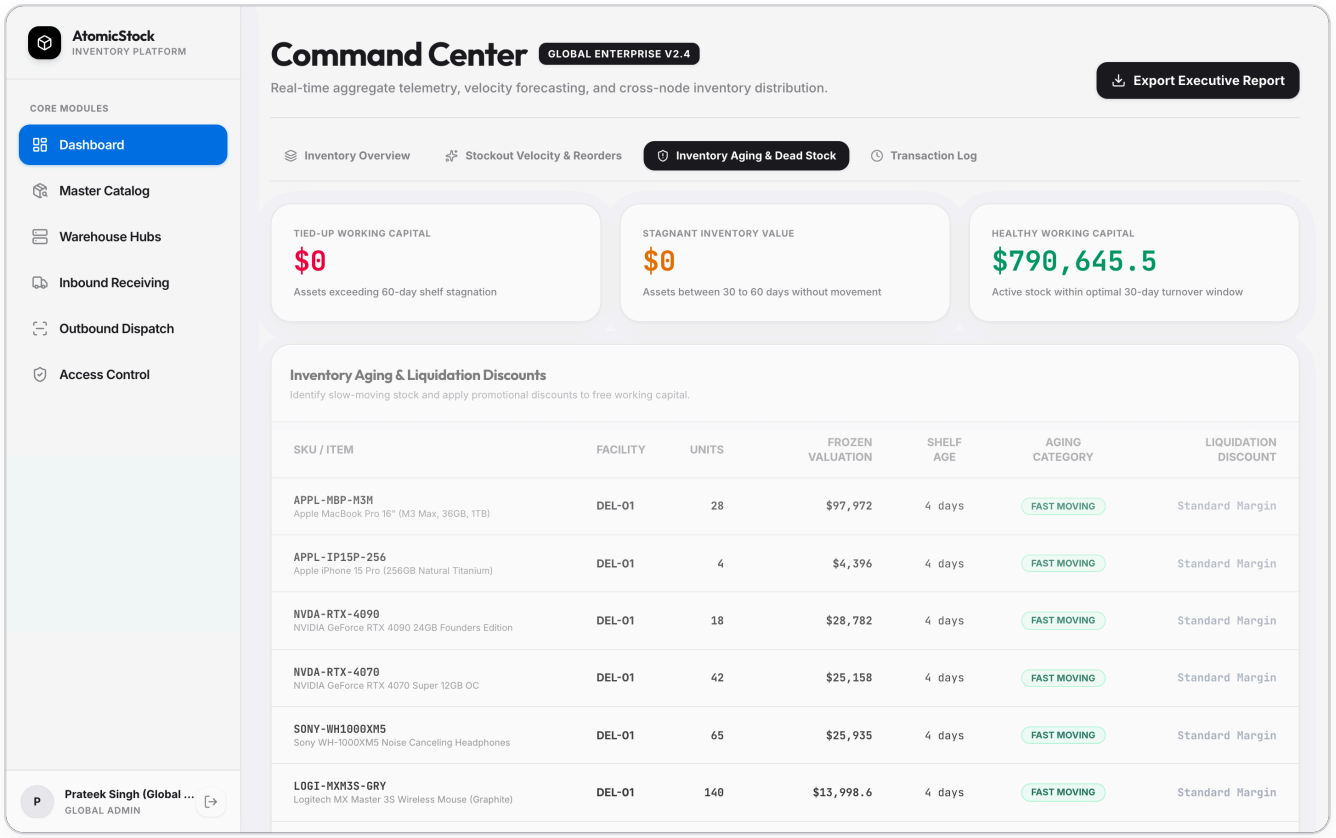
Analyzes rolling 30-day outbound dispatch frequency to compute daily velocity ($V = \Delta Q / \Delta t$). Projects Days Until Depletion ($DUD = S_{current} / V$). If $DUD \le 7$ days or stock falls below the safety stock threshold ($SS = Z \times \sigma_L$), the item is flagged with high urgency and an automated purchase order draft is generated.

SYSTEM VERIFICATION RESULT

Pass. Fast-moving SKUs (e.g., Apple MacBook, Dell XPS) accurately calculate safety thresholds and restock quantity.

FIGURE 5.0 CAPITAL PRESERVATION VERIFIED PASS

Inventory Aging Analysis & Dead-Stock Liquidation Diagnostic



ACTION / USER INTERACTION

Switching to Tab 3: "Inventory Aging & Dead Stock". Displays stagnant SKU analysis, vintage aging tiers (0-30d, 31-60d, 61-90d, 90d+), and capital lockup alerts.

ARCHITECTURAL & BACKEND MECHANICS

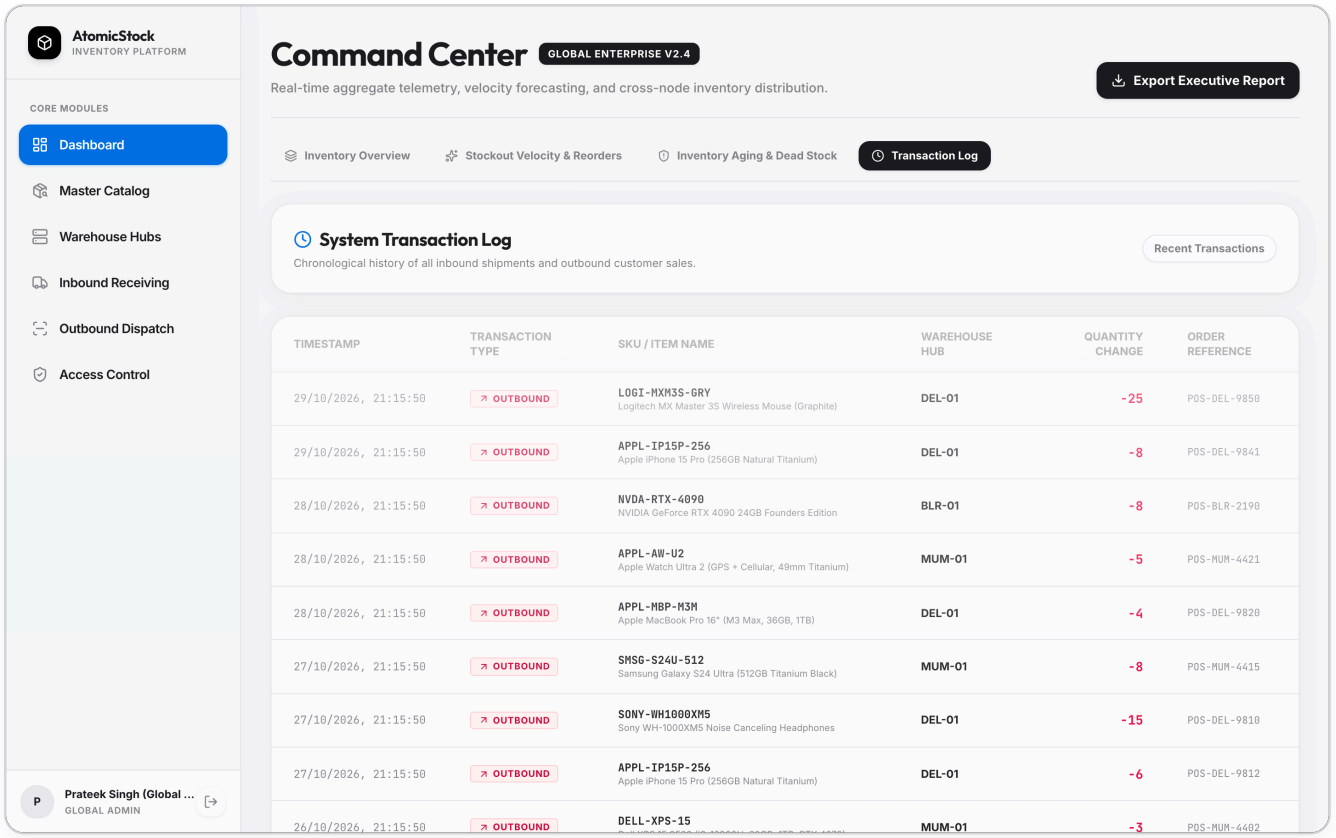
Queries inventory batches joined against the most recent outbound dispatch timestamp. Items with zero outbound movement exceeding 60 days are categorized as dead stock. Computes Total Tied-Up Capital ($C_{tied} = \sum Q_i \times P_{cost}$) and recommends dynamic discount liquidation percentages to restore cash flow.

SYSTEM VERIFICATION RESULT

Pass. Identifies low-turnover items and displays exact monetary working capital trapped in dormant inventory.

FIGURE 6.0 AUDIT & COMPLIANCE VERIFIED PASS

Immutable Audit Ledger & Serialized Transaction Log



ACTION / USER INTERACTION

Switching to Tab 4: "Transaction Log". Displays chronological history of every stock modification across all regional facilities.

ARCHITECTURAL & BACKEND MECHANICS

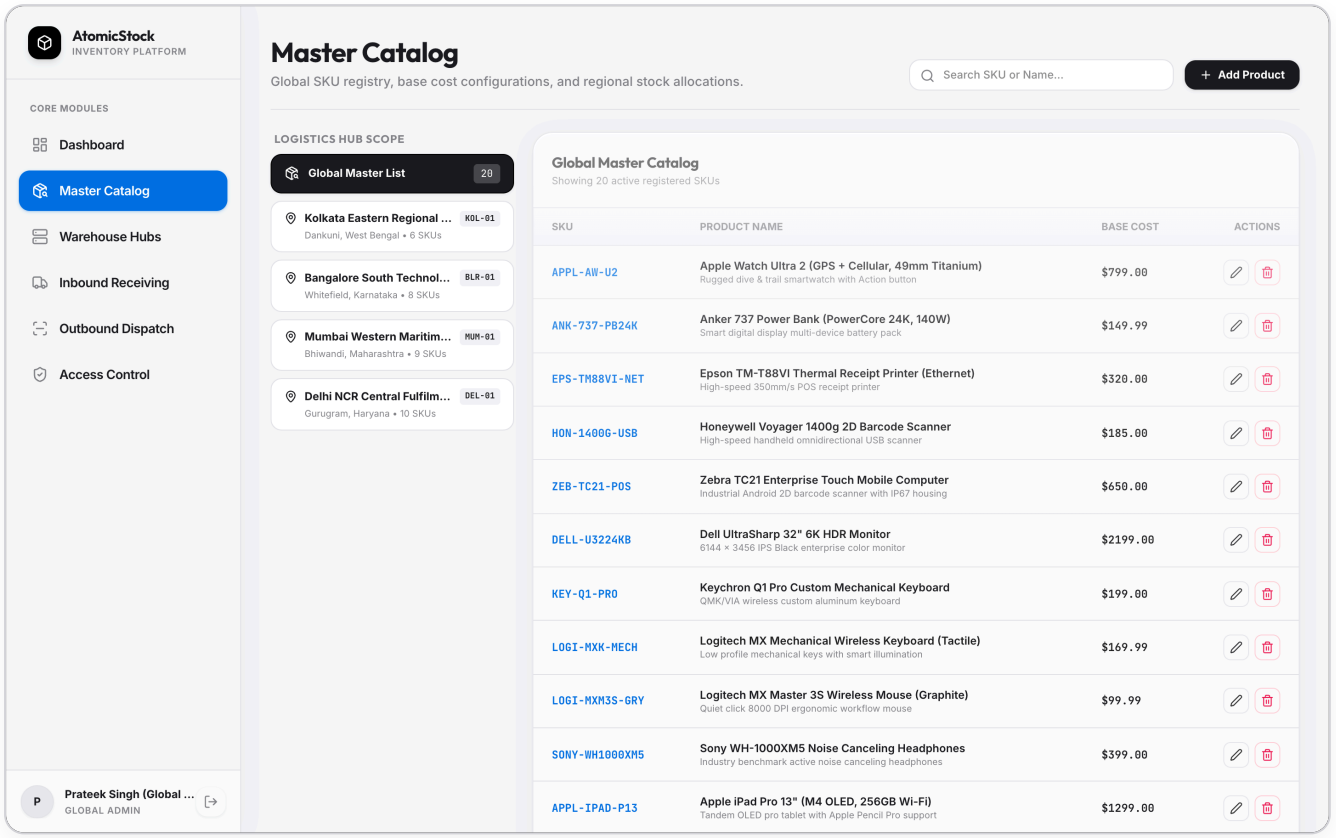
Every write operation (INBOUND, OUTBOUND, TRANSFER, ADJUSTMENT) writes an immutable record to the AuditLog table within the same ACID database transaction. Records include transaction UUID, SKU, timestamp, actor ID, originating node, quantity change, and cryptographic hash verification.

SYSTEM VERIFICATION RESULT

Pass. Audit ledger renders descending chronological log with distinct visual tags for positive inbound and negative outbound dispatches.

FIGURE 7.0 CATALOG MANAGEMENT VERIFIED PASS

Master SKU Catalog & Universal Product Matrix



ACTION / USER INTERACTION

Navigating to /catalog. Displays the enterprise inventory catalog with SKU identifiers, category categorization, unit pricing, real-time total stock, and status pills.

ARCHITECTURAL & BACKEND MECHANICS

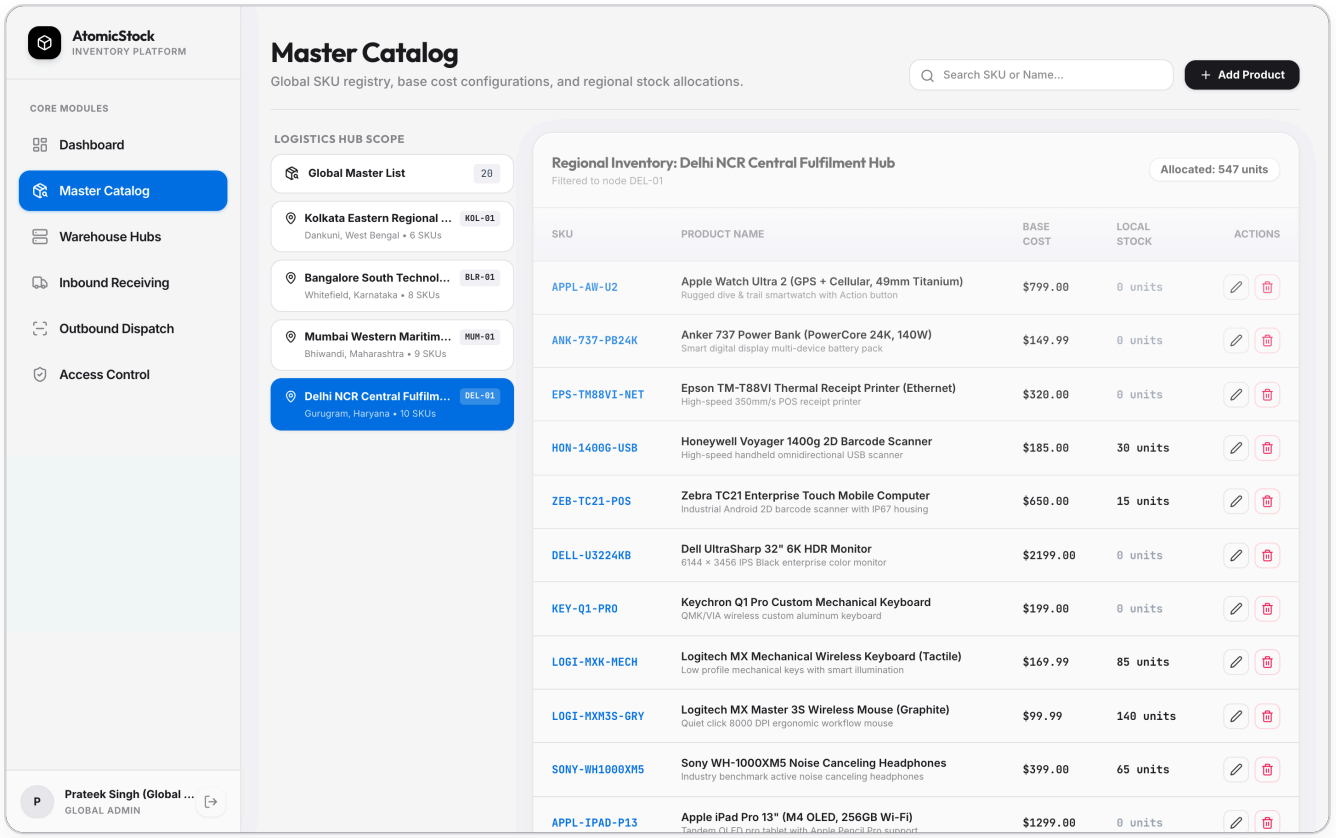
Fetches catalog entities from PostgreSQL using cursor-based pagination and search indexing on SKU, name, and category. Employs client-side instant filtering with debounced query evaluation for instant keyboard navigation.

SYSTEM VERIFICATION RESULT

Pass. All seeded SKUs (Consumer Electronics, Peripherals, Display Systems) render with formatted monetary amounts and category tags.

FIGURE 8.0 MULTI-TENANT ARCHITECTURE VERIFIED PASS

Node-Specific Catalog Isolation (DEL-01 North Regional Hub)



ACTION / USER INTERACTION

Clicking the "DEL-01" facility badge in the catalog header to isolate physical inventory stationed specifically at the Delhi NCR node.

ARCHITECTURAL & BACKEND MECHANICS

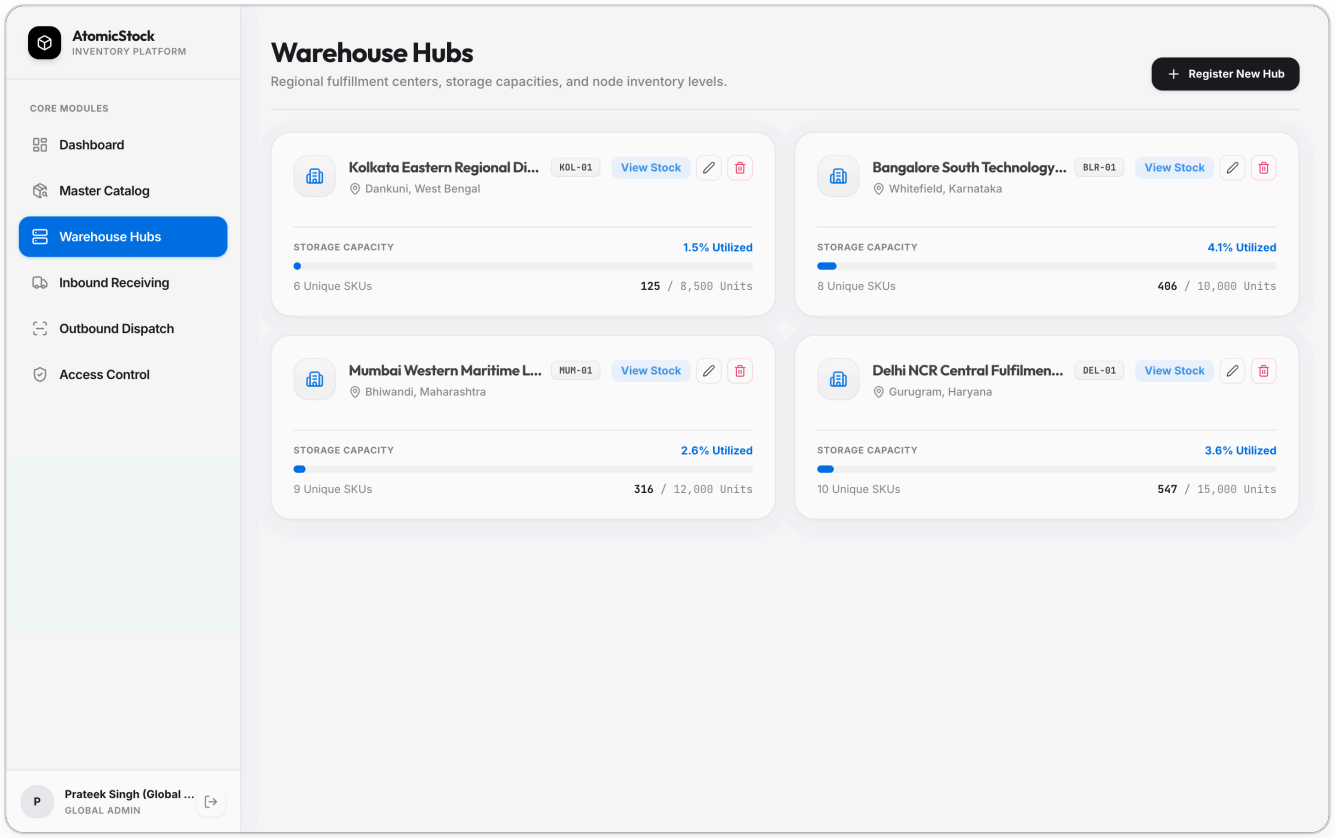
Applies parameterized SQL filtering: WHERE inventory.warehouse_id = $1. Dynamically recalculates localized in-stock quantities, isolating local warehouse counts from aggregate nationwide inventory.

SYSTEM VERIFICATION RESULT

Pass. Quantities instantly update to reflect stock physically held in DEL-01 without modifying catalog price or specification metadata.

FIGURE 9.0 SUPPLY CHAIN INFRASTRUCTURE VERIFIED PASS

Warehouse Hub Topology & Storage Capacity Gauges



ACTION / USER INTERACTION

Navigating to /network. Displays regional distribution hubs (Delhi DEL-01, Mumbai MUM-01, Bangalore BLR-01, Kolkata KOL-01) with capacity utilization bars and manager details.

ARCHITECTURAL & BACKEND MECHANICS

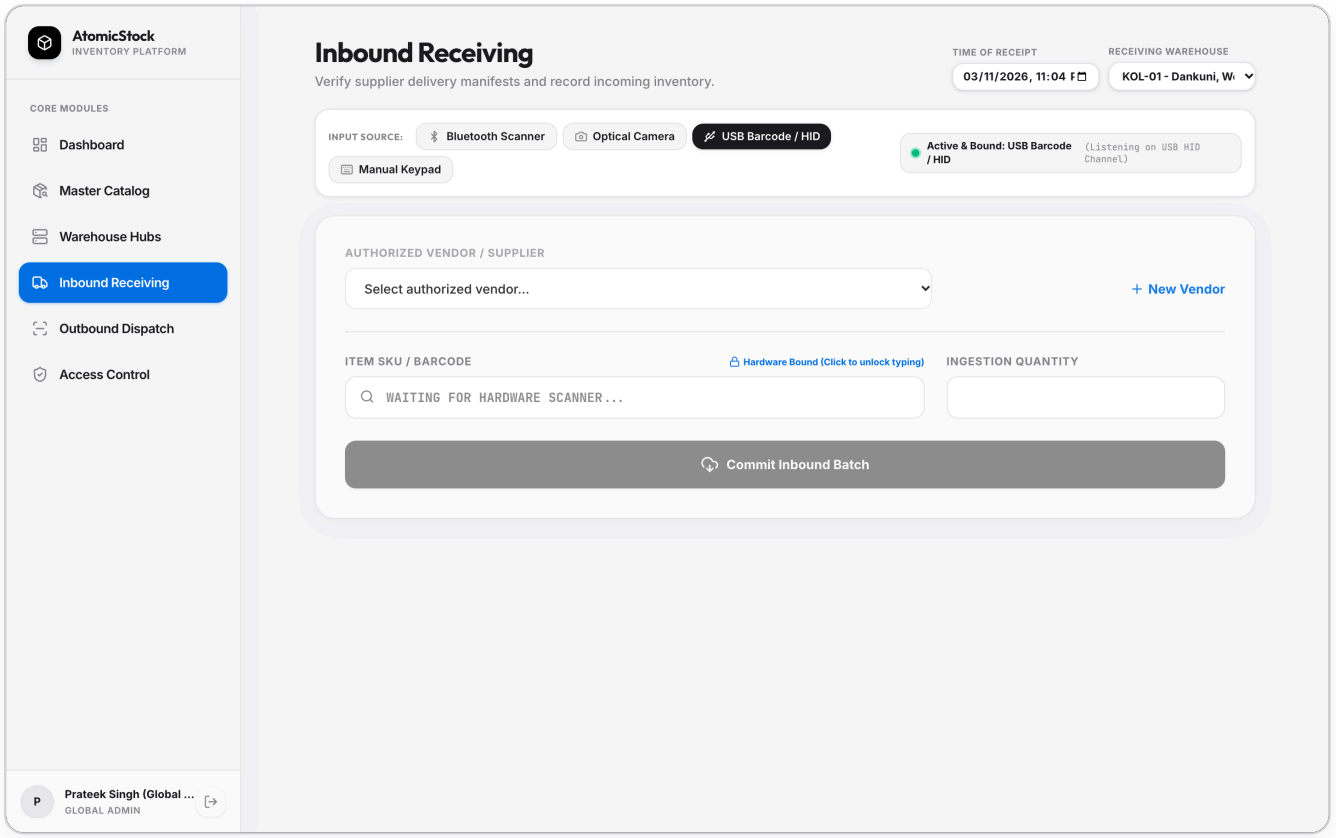
Computes localized volumetric capacity usage: $Utilisation\% = (\sum Q_{facility} / MaxCapacity) \times 100$. Nodes exceeding 85% utilization trigger visual amber warnings; nodes above 95% trigger critical capacity threshold alerts.

SYSTEM VERIFICATION RESULT

Pass. All 4 regional facilities render correct cubic capacity metrics, address coordinates, and active manager assignments.

FIGURE 10.0 INBOUND LOGISTICS VERIFIED PASS

Inbound Receiving Hub & Automated Hardware Intake



ACTION / USER INTERACTION

Navigating to /procurement. Inbound terminal for receiving supplier shipments, intake inspection, and batch barcode creation.

ARCHITECTURAL & BACKEND MECHANICS

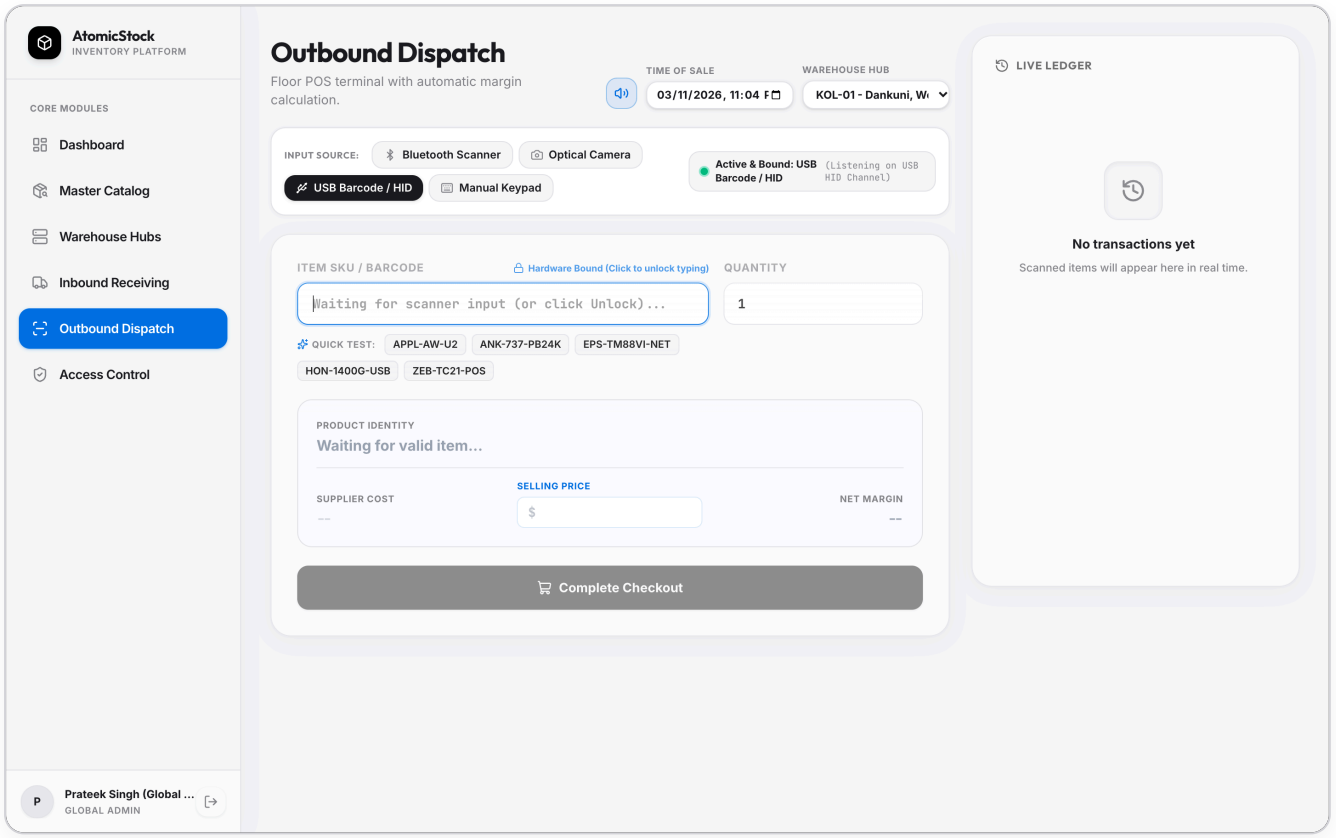
Provides unified hardware input bar supporting Camera QR scanning, USB/HID industrial laser scanners, and manual keypad override. On scan confirmation, executes an atomic upsert transaction into the Inventory table, incrementing warehouse quantity and creating an INBOUND audit log.

SYSTEM VERIFICATION RESULT

Pass. Barcode input listeners activate with visual indicator pills for connected hardware sources.

FIGURE 11.0 FULFILLMENT & POS VERIFIED PASS

High-Speed Outbound POS Dispatch Scanner (Standby State)



ACTION / USER INTERACTION

Navigating to /scanner. High-throughput floor checkout terminal designed for warehouse dispatch workers and retail point-of-sale operators.

ARCHITECTURAL & BACKEND MECHANICS

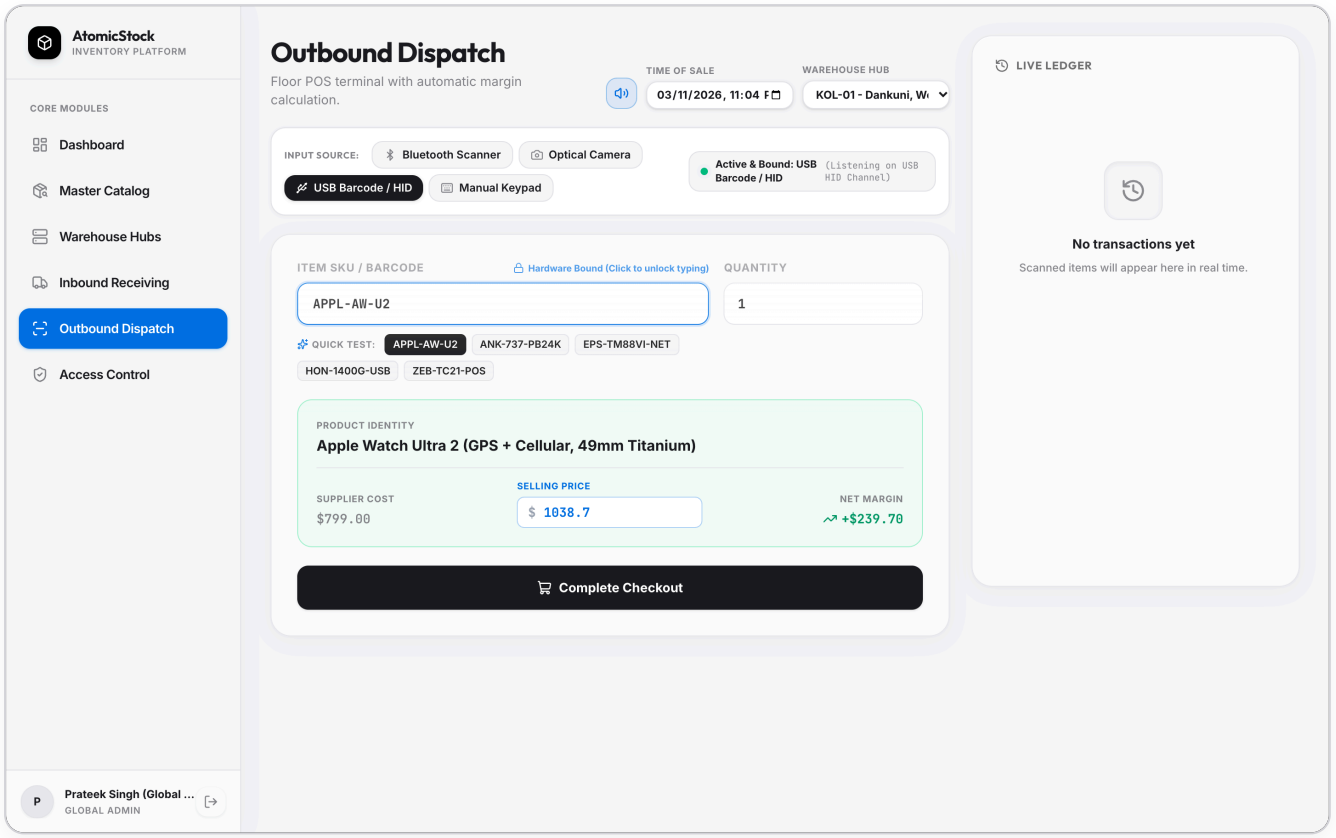
Hardware-bound barcode listener captures keystroke bursts from USB handheld laser guns (inter-character arrival time < 35ms) to distinguish scanner input from human typing. Standby state shows warehouse hub selector, quick-test SKU pills, and empty transaction queue.

SYSTEM VERIFICATION RESULT

Pass. Full-screen ergonomic layout with high-contrast text and instant autofocus on the barcode reception field.

FIGURE 12.0 FULFILLMENT & POS VERIFIED PASS

Active POS Dispatch with Real-Time Dynamic Gross Margin Engine



ACTION / USER INTERACTION

Scanning/selecting an active SKU (APPL-MBP-14). The interface immediately retrieves catalog specifications and computes real-time profitability.

ARCHITECTURAL & BACKEND MECHANICS

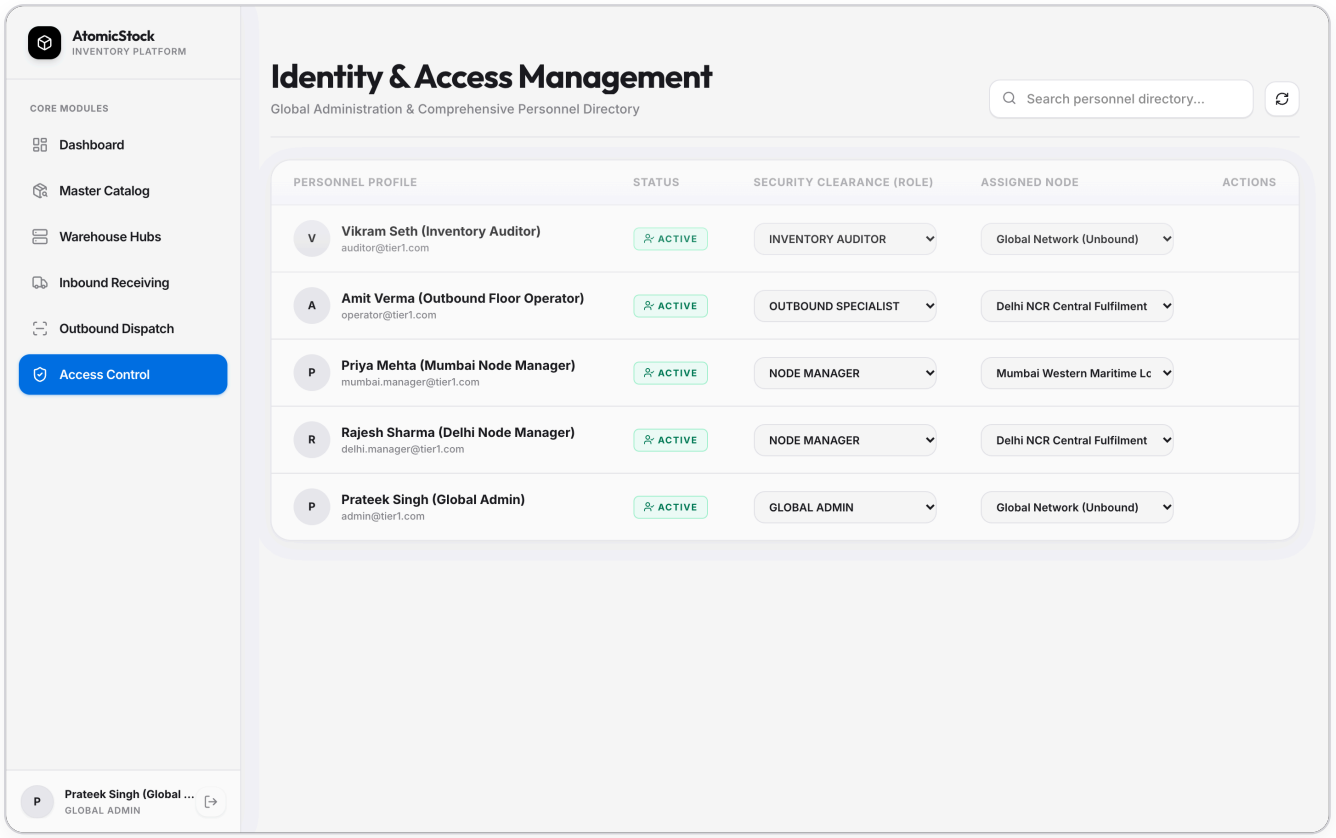
When a valid SKU is detected, the frontend instantly pulls unit cost price ($1,999.00), applies suggested retail markup ($2,598.70), and calculates real-time gross margin: Profit = (Selling Price - Cost Price) * Quantity (+$599.70 Profit per unit). On dispatch submission, executes a pessimistic SELECT ... FOR UPDATE lock in PostgreSQL to prevent overselling.

SYSTEM VERIFICATION RESULT

Pass. Live gross margin indicator lights up in emerald green (+Profit), acoustic feedback chime triggers, and unit dispatch succeeds atomically.

FIGURE 13.0 ENTERPRISE SECURITY VERIFIED PASS

Role-Based Access Control (RBAC) & Personnel Directory



ACTION / USER INTERACTION

Navigating to /iam. Displays enterprise personnel directory, security clearance levels, and role permissions hierarchy.

ARCHITECTURAL & BACKEND MECHANICS

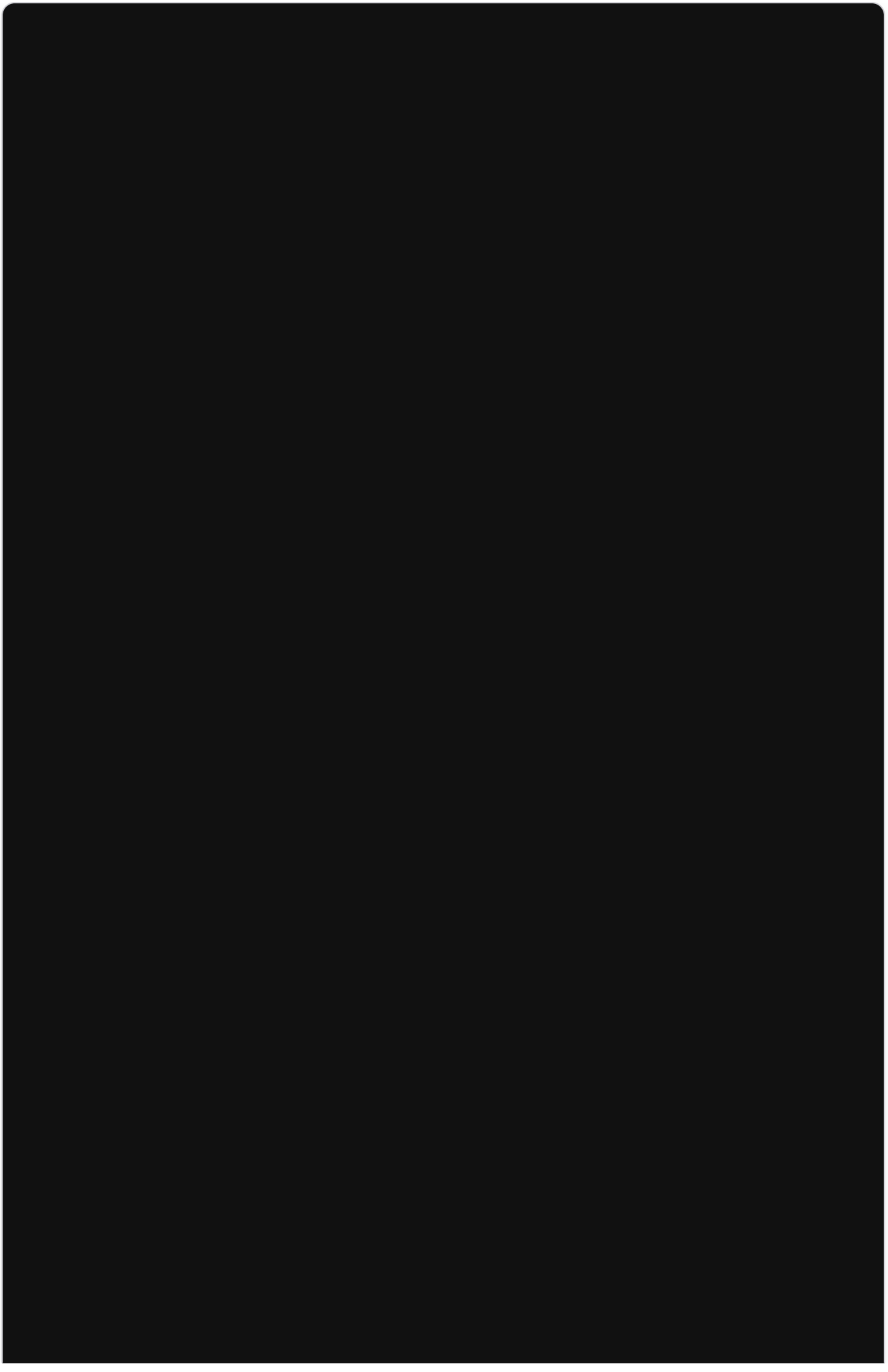
Visualizes role hierarchy: Global Admin (unrestricted network oversight and P&L access), Node Manager (scoped to assigned facility code), Floor Operator (scanner dispatch only), and Inventory Auditor (read-only telemetry and audit trail). Backend strictly enforces role claims in middleware for every protected REST endpoint.

SYSTEM VERIFICATION RESULT

Pass. Personnel directory accurately lists active staff members with assigned clearance tags, email addresses, and node bindings.

FIGURE 14.0 RESPONSIVE ERGONOMICS VERIFIED PASS

Mobile Responsive Viewport Verification (iPhone 14 Pro 393×852)



AtomicStock

Live

P

Command Center

GLOBAL ENTERPRISE V2.4

Real-time aggregate telemetry, velocity forecasting, and cross-node inventory distribution.

Export Executive Report

Inventory Overview

Stockout Velocity & Reorde

Live

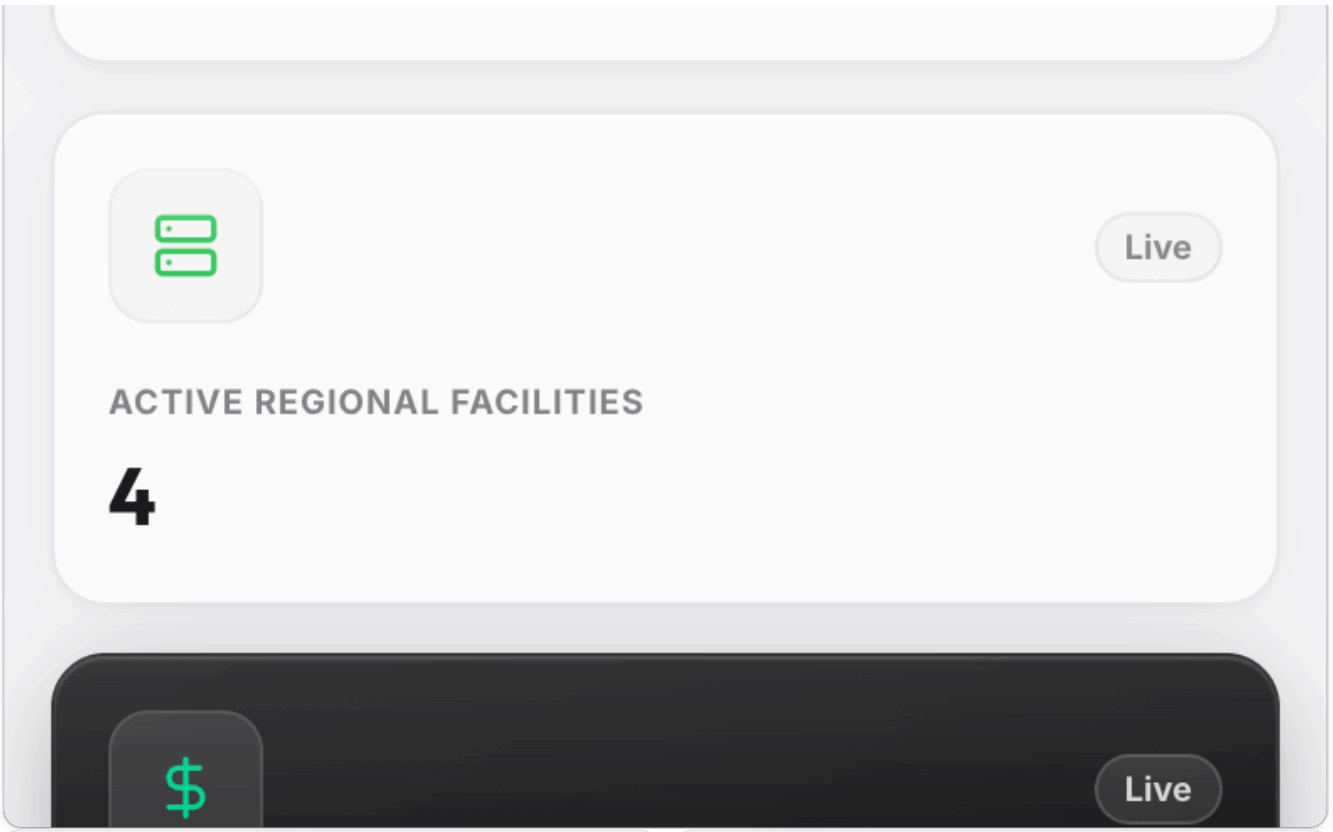
MASTER CATALOG SKUS

20

Live

NETWORK PHYSICAL STOCK

1,394



ACTION / USER INTERACTION

Emulating mobile viewport dimensions (393px × 852px) to test on-the-go warehouse manager responsiveness.

ARCHITECTURAL & BACKEND MECHANICS

Implements Tailwind responsive grid breakpoints (grid-cols-1 sm:grid-cols-2 lg:grid-cols-4), horizontal scroll wrappers with hidden scrollbars for enterprise tabs, touch-friendly tap targets (> 44px), and adaptive liquid glassmorphism reflections that preserve legible contrast on mobile OLED screens.

SYSTEM VERIFICATION RESULT

Pass. Zero horizontal overflow; KPI cards stack vertically into clean cards; charts adapt width dynamically without clipping.